



Novel hybrid D/A structures for high-resolution SAR ADCs—analysis, modeling and realization

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ABSTRACT

An intermittent-sleeping C–R hybrid D/A structure, a resistor-reusing R–C hybrid D/A structure and an R–C–R combined D/A structure are researched with theoretical analysis and Matlab modeling in this paper. The switching power and passive components' matching requirement of these D/A structures are discussed in detail. Three SAR ADCs with these three novel structures are realized based on SMIC 0.18 μm CMOS, SMIC 90 nm CMOS and UMC 90 nm CMOS, respectively. The theoretical analysis, modeling verification and circuit realization proves the applicability of these three D/A conversion networks to high-resolution SAR ADCs.

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1. Introduction

As the significant building blocks in SoC's, A/D converters have been widely used in many applications such as HDTV, GSM, UWB communication and so on [1–12]. SAR A/D converters are characterized by simple architectures, small areas and low power, making them especially popular for touch screen controllers, wireless sensor networks and biomedical instrumentations [6–13].

Either capacitor array or resistor string can be used to implement the internal D/A conversion network, which is the pivotal building block of SAR ADC [12,14]. But with the resolution increasing, the total number of the passive components in these two types of D/A networks will increase exponentially, making them not very suitable for high-resolution applications if the trade-off between small area and good matching performance is considered. To reduce the chip size, small-area passive components should be used, which will deteriorate the matching performance of the D/A network and the linearity of the ADC. Though additional calibration techniques can be utilized to guarantee the converter's linearity, it will increase the total power dissipation and circuit complexity [10,14]. C–R and R–C hybrid D/A networks are usually used in high-resolution SAR ADCs (≥ 12 -bit). But in the traditional hybrid structures, the resistor strings will consume much static power, and keeping the capacitors

in stringent binary-ratio will cost much chip area and layout routing task [7,11,12,14]. Further research on high-resolution D/A conversion networks is still very necessary for SAR ADC applications.

In this paper, three hybrid D/A conversion structures for SAR ADCs are discussed in detail. Using the intermittent-sleeping C–R hybrid D/A structure, more than half of the power consumed by the resistor string can be reduced. In the resistor-reusing R–C hybrid D/A structure, the MSBs' (Most Significant Bits) resistor string is reused by a binary-ratio capacitor pair. Compared with traditional charge redistribution and voltage division structures, the chip area is reduced. And only two capacitors need to be matched well, making the layout routing more flexible than traditional hybrid structures. In the R–C–R combined D/A structure, the matching requirement of the capacitor array is alleviated by utilizing a two-segment reusable resistor string. Compared with traditional C–R structure, the size and switching power of the capacitor array in this structure can both be dramatically reduced. The applicability of these three hybrid D/A conversion structures to high-resolution SAR ADCs are also proved by circuit design and realization in this paper.

2. D/A conversion networks for SAR ADCs

The SAR ADC architecture can be illustrated by Fig. 1. It consists of an S/H (Sample/Hold) circuit, a comparator, a SAR control unit and a D/A conversion network. Its operation is based on a “binary search” algorithm. The analog input is sampled and

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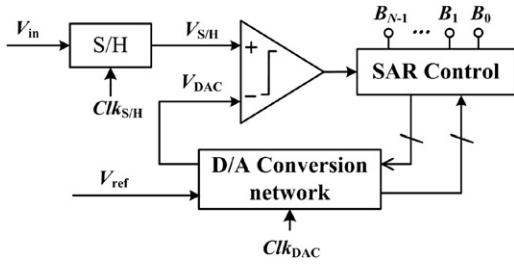


Fig. 1. Architecture of SAR A/D converter.

compared with the sub-references produced by the D/A converter, which is under the control of the logic circuit. The sub-reference at particular comparison stage depends on the value of the calculated bits. During the decision of B_k ($0 \leq k \leq N-1$, and $B_{N-1} \sim B_{k+1}$ have been generated before the decision of B_k), assume $B_k = "1"$ at first, and then, the sub-reference voltage can be expressed by: $V_{DAC} = B_{N-1}V_{ref}/2^1 + B_{N-2}V_{ref}/2^2 + \dots + 1 \times V_{ref}/2^{N-k}$, where N is the resolution of the ADC and V_{ref} represents the full-scale reference voltage of the converter. If the sample-and-hold value $V_{S/H}$ is less than the sub-reference voltage V_{DAC} , B_k should be "0", otherwise, $B_k = "1"$. All the digital outputs $B_{N-1} \sim B_0$ can be sequentially generated based on this operation.

As the pivotal building block of SAR ADC, D/A conversion network generates the sub-reference voltages, which are used to compare with the sampled input for the generation of the digital outputs. The linearity, switching power and operation speed of the internal D/A converter can affect the performance of the SAR ADC directly. Especially in high-resolution SAR ADC, the D/A conversion network usually occupies the most area of the whole chip, and its matching performance is the key factor that influences the linearity of the SAR ADC.

In this section, three hybrid D/A conversion structures suitable for high-resolution SAR ADC applications are analyzed theoretically and verified by modeling using Matlab. The matching performance requirement and the switching power of these D/A conversion networks are discussed.

2.1. Intermittent-sleeping C–R hybrid structure

The structure and operation of traditional C–R hybrid D/A conversion network of SAR ADC can be seen in Refs. [7] and [14]. During its operation, much static power will be consumed by the resistor string. With regard to this shortcoming, an intermittent-sleeping approach is proposed. Fig. 2 shows a 10-bit intermittent-sleeping D/A conversion structure, where a logic controlled switch S_r is introduced. During the sampling step and the 5MSBs' generation, the resistor string is in the power-down mode with S_r turned off. After the 5 MSBs are generated, the switch S_r is turned on and then, the resistor string is used to generate the 5 LSBs (Least Significant Bits). The control signal of S_r has been shown in Fig. 2. Obviously, using this approach, more than half of the static power consumed by the resistor string can be reduced.

In SAR ADC design, it is important to consider the passive components' matching performance. With the analysis approach proposed in Ref. [5] and assuming that the capacitor values are normally distributed with mean C_0 and standard deviation σ_c , then if 5-bit capacitor array is used to generate the 5 MSBs of a 10-bit ADC, σ_c/C_0 should be no more than $2^{(5-1)/2-10}$ to guarantee INL (Integral Nonlinearity) < 0.5 LSB. Based on this discussion, conditions of $\sigma_c/C_0 = 2^{-7.5}$ and $\sigma_c/C_0 = 2^{-8.5}$ are both verified by Matlab models. Shown in Fig. 3 is the verification results, which can prove that σ_c/C_0 should better be less than 2^{-8} . With 1.8 V reference voltage, the energy dissipation of the capacitor array is shown in Fig. 4.

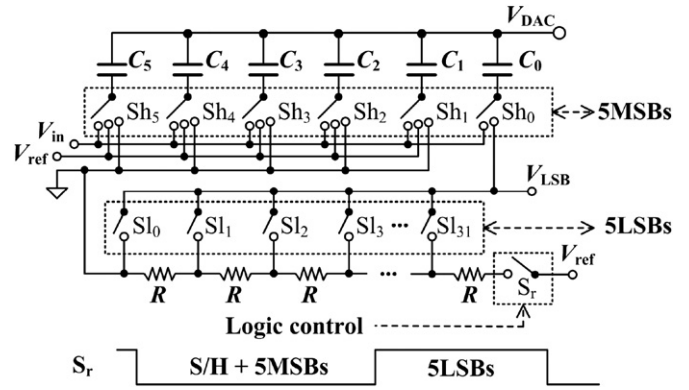


Fig. 2. Intermittent-sleeping C–R hybrid D/A structure for SAR ADC.

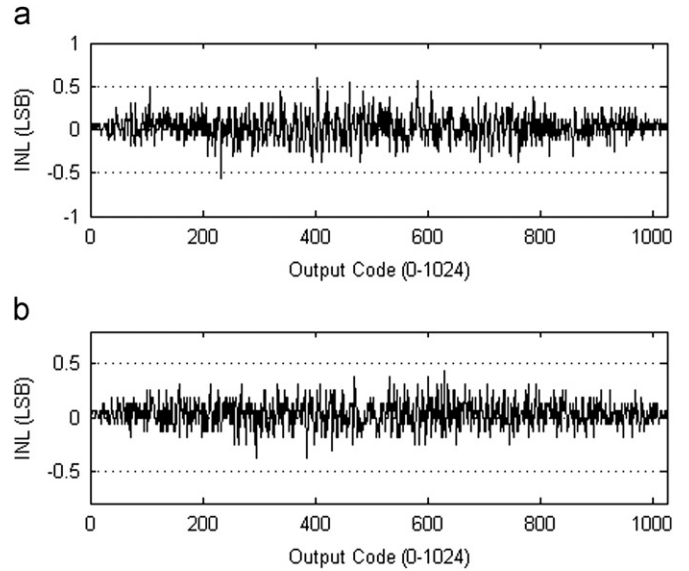


Fig. 3. Capacitor's mismatch (a) $\sigma_c/C_0 = 2^{-7.5}$ (b) $\sigma_c/C_0 = 2^{-8.5}$.

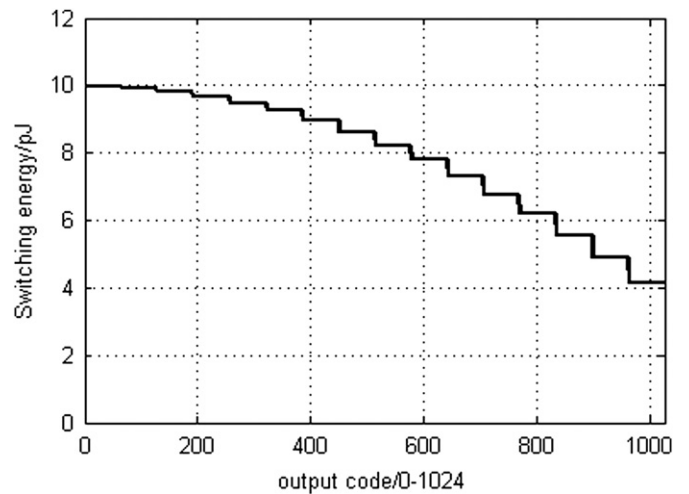


Fig. 4. Switching energy of the 5-bit capacitor array.

The total power of the D/A conversion network is mainly composed of the capacitors' dynamic power and the resistor string's static power. For 200 kS/s sampling rate, the conversion period is 5 μ s. Thus, from Fig. 4, it can be seen that the maximum power dissipation of the capacitor array is only about 2 μ W.

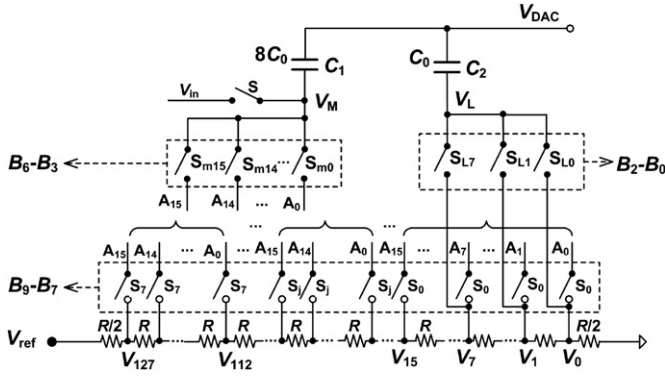


Fig. 5. Resistor-reusing R-C hybrid D/A structure for SAR ADC.

It appears at the smallest output code, where only down transitions happen. The resistor string consumes the most power of the whole D/A conversion. But in this intermittent-sleeping approach, the resistor string is on the power-down state during sampling and generating the 5 MSBs, saving more than half of the reference power.

2.2. Resistor-reusing R-C hybrid structure [12]

In the charge redistribution structure and the traditional hybrid structure, the capacitors should be stringently binary-ratio, which makes that the capacitors should be carefully routed in the layout design. Shown in Fig. 5 is a novel “7 MSBs+3 LSBs” R-C hybrid structure where 10-bit resolution is realized with a 7-bit resistor string reused by an 8:1 capacitor pair. Compared with traditional charge redistribution and voltage division structures, the chip area is reduced. Since only two capacitors need to be matched well, the layout routing can be more flexible than traditional structures. And the capacitors’ inherent sample-and-hold function makes that no additional S/H circuit is needed in this structure. The operation of this D/A conversion structure can be seen in [12].

In the resistor-reusing R-C hybrid structure, the 7-bit resistor string should be matched well to satisfy the linearity requirement of a 10-bit ADC. Similarly, with the analysis approach proposed in Ref. [5] and assuming that the resistor values are normally distributed with mean R_0 and standard deviation σ_R , σ_R/R_0 should be not more than $2^{(7-1)/2-10}$. Fig. 6 gives the Matlab verification results under the conditions of $\sigma_R/R_0=2^{-7.5}$ and $\sigma_R/R_0=2^{-6.5}$, respectively, proving that σ_R/R_0 should better be less than 2^{-7} . Since the accuracy of the capacitor pair’s ratio can only affect the 3-bit LSBs in this ADC, the matching requirement for them is not very stringent. Under the condition of 3.3 V reference voltage, the energy dissipation of the capacitors is shown in Fig. 7. The capacitors should be connected to higher voltages than before during the up transitions, so more switching energy is dissipated at larger output codes.

For 2.5 MS/s sampling rate, the conversion period is 0.4 μ s. Thus, from Fig. 7, it can be seen that the maximum power dissipation of the capacitors is only about 7.5 μ W. It appears at the largest output code, where only up transitions happen. The static power of the resistor string can be calculated as $P_{\text{static}}=V_{\text{ref}}^2/R_{\text{total}}$, where V_{ref} is the reference voltage and R_{total} represents the total resistor.

2.3. R-C-R combined D/A structure

In traditional C-R hybrid D/A structure, the capacitors are used to generate the MSBs, so they must be matched very well. Without calibration techniques, large-area capacitors should be

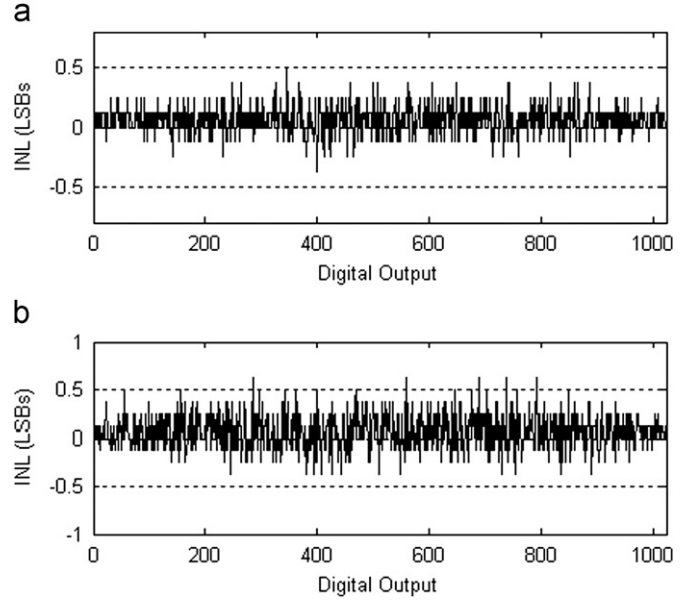


Fig. 6. Resistors' mismatch (a) $\sigma_R/R_0=1/2^{7.5}$ (b) $\sigma_R/R_0=1/2^{6.5}$.

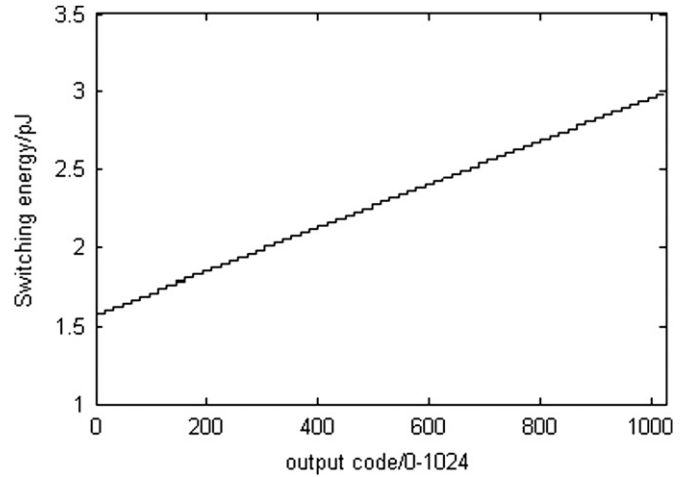


Fig. 7. Switching energy of the 8:1 capacitor pair.

used to satisfy the matching requirement, making the chip area very large. Though calibration techniques can be used to solve the nonlinearity problem caused by capacitors’ mismatch, the circuit design will be much more complex and much power will be consumed.

In this section, a novel R-C-R combined D/A structure is presented, where a two-segment reusable resistor string is utilized to alleviate the capacitors’ matching requirement. Fig. 8 shows a 12-bit R-C-R combined D/A conversion network, which is composed of a 6-bit two-segment reusable resistor string and a 6-bit capacitor array. In the sample phase, the input signal V_{in} is sampled by the capacitor array with switch S_a turned on. Then, the converter enters the hold step. The switch S_a will be turned off, and the bottom plates of the capacitor array are connected to V_{cm} with switch S_{DM} turned on. V_{cm} is generated by the resistor string ($V_{\text{cm}} \approx (V_{\text{top}} - V_{\text{bot}})/2$), as shown in Fig. 8. In the hold step, $V_{\text{DAC}}=2V_{\text{cm}} - V_{\text{in}}$. From the comparison of this value with V_{cm} , the MSB B_{11} can be generated in the hold step. If $V_{\text{DAC}} < V_{\text{cm}}$, $B_{11} = “0”$, otherwise, $B_{11} = “1”$. Next, the converter enters the approximation step. Based on the result of B_{11} , V_r is connected to the different parts of the resistor string with the switch S_r connected to A or B. If $B_{11} = “1”$, the top resistor string ($V_{\text{top}} \sim V_{\text{cm}}$) is selected with S_r

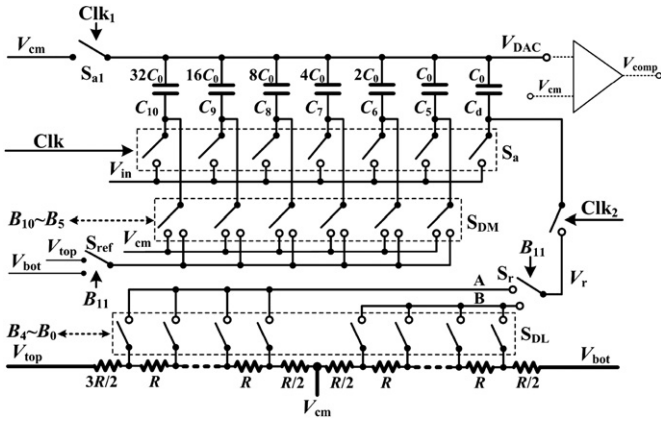


Fig. 8. R-C-R combined D/A structure for SAR ADC.

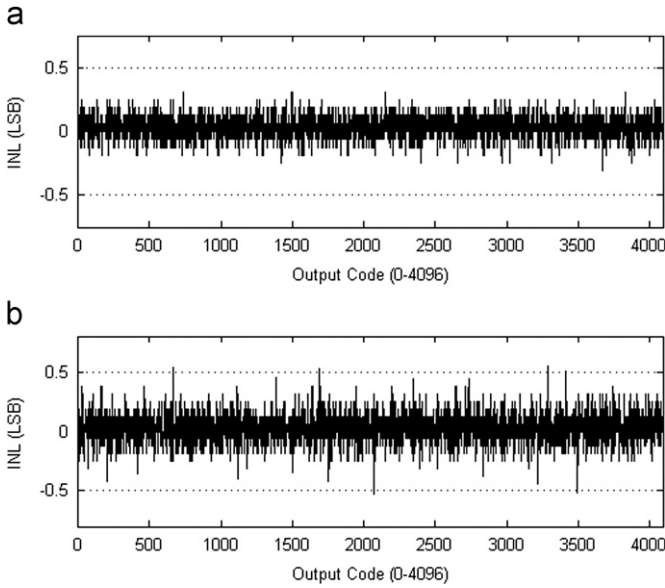


Fig. 9. Capacitor's mismatch (a) $\sigma_c/C_0=2^{-8}$ (b) $\sigma_c/C_0=2^{-9}$.

connected to A, thus, V_{top} and V_{cm} are used as the positive and negative reference, respectively, during the following D/A conversion. Otherwise, S_r is connected to B, making V_{cm} and V_{bot} as the positive and negative reference, respectively, during the next operation. Based on the binary search operation similar as that of the traditional C-R hybrid structure, the digital outputs $B_{10} \sim B_0$ can be produced. $B_{10} \sim B_5$ are generated by the capacitor array while $B_4 \sim B_0$ are generated by reusing the two-segment resistor string.

In this R-C-R combined D/A structure, the matching requirement for the capacitor array is relaxed compared with traditional C-R hybrid structure. Since the MSB B_{11} is generated directly in the hold step, the 6-bit capacitor array need to be matched to satisfy just 11-bit linearity requirement, which means that 1-bit requirement has been relaxed. From Ref. [5], if the capacitor values are normally distributed with mean C_0 and standard deviation σ_c , then σ_c/C_0 should be not more than $2^{-8.5}$. Fig. 9 shows the verification results using Matlab.

Since 1-bit matching requirement for the capacitor array has been relaxed by reusing the two-segment resistor string, the unit capacitor can be half of that in the traditional "6 MSBs+6 LSBs" C-R hybrid D/A structure, making the switching energy and the area of the capacitor array dramatically reduced. Furthermore, during the operation of the capacitor array when generating

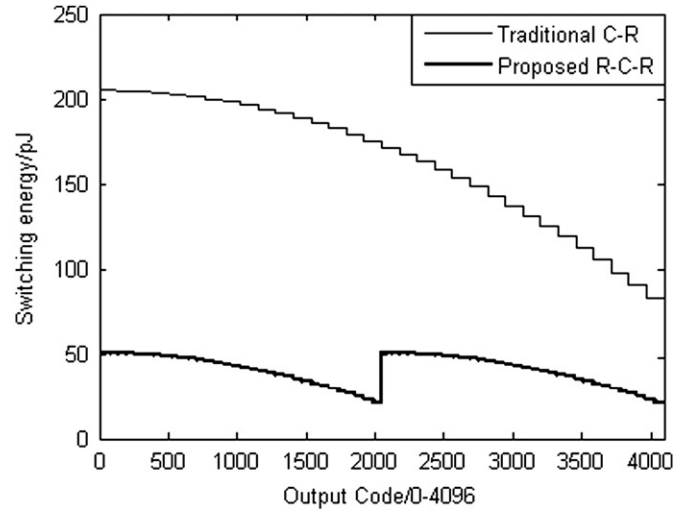


Fig. 10. Switching energy of the 6-bit capacitor array.

$B_{10} \sim B_5$, the difference between positive reference and negative reference has been halved compared with traditional D/A structures, making the switching energy reduced further. Under the condition of 3.3 V reference voltage, the switching energy of this capacitor array and that in traditional "6 MSBs+6 LSBs" C-R structure has been compared in Fig. 10.

3. Design results and discussion

With these high-resolution D/A conversion structures discussed in Section 2, three SAR ADC IP cores are realized in SMIC 0.18 μm CMOS, SMIC 90 nm CMOS and UMC 90 nm CMOS, respectively. In this section, the circuit implementation and the measurement results of the three SAR ADCs are described and analyzed.

3.1. A 10-bit 0.18 μm C-R hybrid SAR ADC

With the intermittent-sleeping C-R hybrid D/A structure discussed in Section 2.1, an 8-channel 10-bit 200 kS/s SAR ADC IP core is realized in SMIC 0.18 μm CMOS 1P5M process. Shown in Fig. 11 is the 10-bit 0.18 μm SAR ADC layout photo. The active area of this IP core is $333 \mu\text{m} \times 250 \mu\text{m}$. For improving the technology compatibility, metal finger capacitors are adopted to form the capacitor array. With the consideration of the matching performance and the power dissipation of the resistor string, the P+ doped poly-silicon resistors are utilized. To achieve a good performance, the resistor string is sided by dummies to maintain the same environment on both sides, and the capacitor array is routed with an axial symmetrical layout topology and dummy-surrounding method to reduce the mismatch. In this A/D converter, 120 fF unit capacitors and 4 k Ω unit resistors are used.

The DC characteristics of this converter are measured with the histogram testing method [15]. With 1.8 V power supply and about 57.2 kHz input, the DNL and INL of this converter are measured to be about 0.32 and 0.81 LSB, respectively, as shown in Fig. 12. During the AC performance measurement, the non-coherent sampling method is utilized [16]. And before DFT, windowing technique is used to avoid the spectral leakage [17]. Shown in Fig. 13 is an 8000 points DFT spectrum for a 91 kHz single-ended sinusoidal input signal sampled at 200 kS/s. The ENOB is 9.15. The SFDR and THD are measured to be 63.2 and 62.7 dB, respectively. Fig. 14 shows the variation of the ENOB

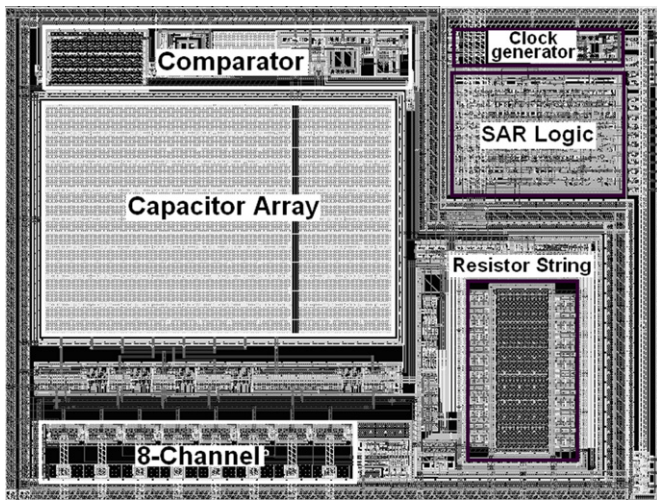


Fig. 11. Layout photo of the 10-bit 0.18 μm SAR ADC.

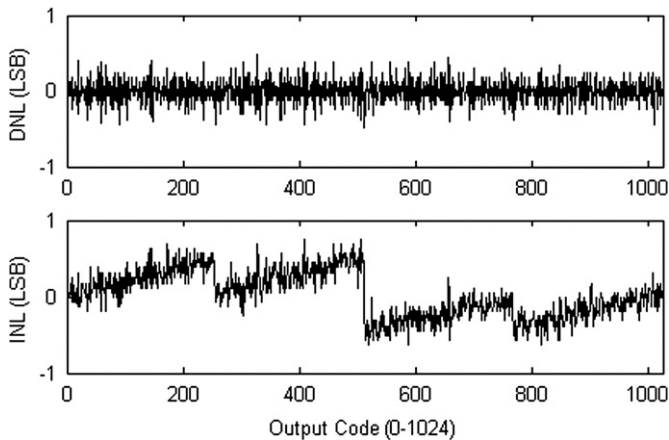


Fig. 12. Nonlinearity of the 10-bit 0.18 μm SAR ADC.

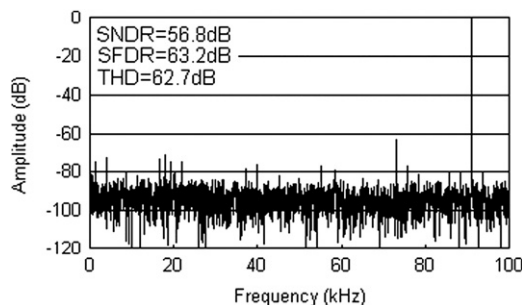


Fig. 13. 8k points DFT @ 200 kS/s with 91 kHz input.

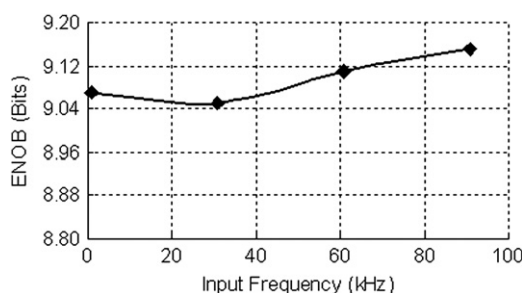


Fig. 14. ENOB @ 200 kS/s vs. input frequency.

with the input frequency. Including the output drivers, the power dissipation of this SAR ADC is 136 μW .

3.2. A 10-bit 90 nm R-C Hybrid SAR ADC

Based on SMIC 90 nm CMOS process, an 8-channel 10-bit 2.5 MS/s SAR ADC IP core is realized with a “7 MSBs+3 LSBs” resistor-reusing R-C hybrid D/A conversion structure [6]. The layout photograph of this converter is shown in Fig. 15, where it just occupies an area of $238 \mu\text{m} \times 214 \mu\text{m}$. For the layout design, the P+ diffusion resistors and finger-cap capacitors are used. Similar as that in the 10-bit 0.18 μm SAR ADC realized in Section 3.1, the resistor ladder is sided by dummies to maintain the same environment on both sides, and the capacitor pair is routed with a symmetry topology and dummy-surrounding method. In this 2.5 MS/s SAR ADC, 60 Ω unit resistors and 120 fF unit capacitors are used to satisfy the matching, speed and power requirement.

Fig. 16 shows measured nonlinearity of the 10-bit 90 nm R-C hybrid SAR ADC. Using the histogram testing method, with an input signal of about 301 kHz sampled at 2.5 MS/s, about 2 M datum are sampled and processed. The DNL and INL are measured

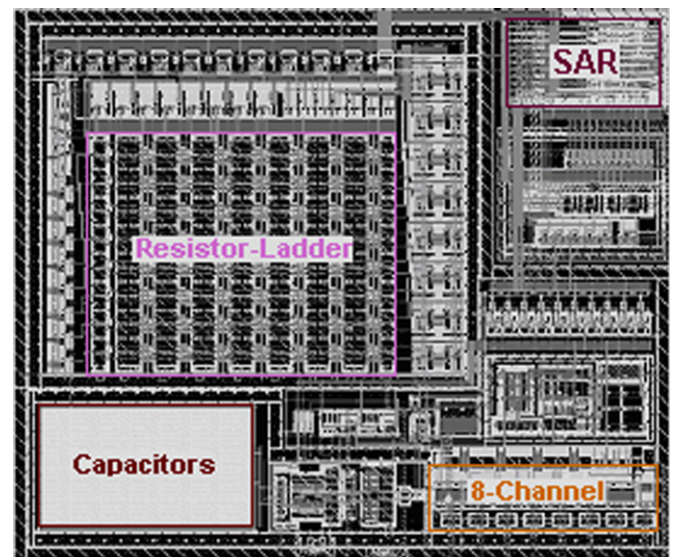


Fig. 15. Layout photo of the 10-bit 90 nm SAR ADC.

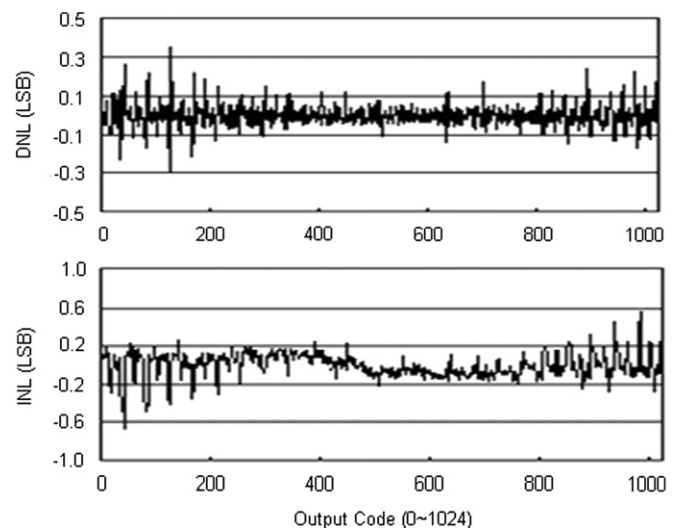


Fig. 16. Nonlinearity of the 10-bit 90 nm SAR ADC.

to be 0.36 and 0.69 LSB, respectively. The AC characteristics are measured at 2.5 MS/s sampling rate. Shown in Fig. 17 is a 5000 points DFT spectrum for a 100 kHz single-ended sine input signal sampled at 2.5 MS/s. The non-coherent sampling method and windowing technique is utilized during the measurement of the AC performance. The measured ENOB is 9.41. The SFDR and THD are measured to be 65.02 and 61.92 dB, respectively. The graphic with ENOB of this 2.5 MS/s ADC versus the input frequency is given in Fig. 18. In the Nyquist input bandwidth, the ENOB is above 9.26. The typical power dissipation is measured to be 6.62 mW including the output drivers.

3.3. A 12-bit 90 nm R–C–R combined SAR ADC

The R–C–R combined D/A structure is validated by the realization of a 12-bit SAR ADC IP core in UMC 90 nm CMOS. Shown in Fig. 19 is the microphotograph of this converter, where the I/O

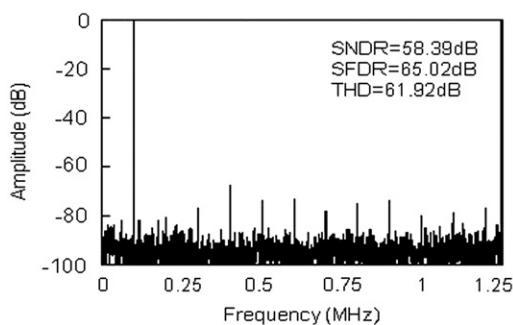


Fig. 17. 5k points DFT @ 2.5 MS/s with 100 kHz input.

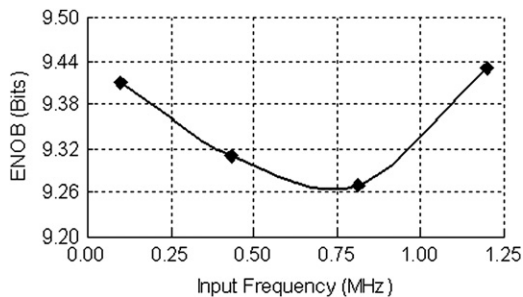


Fig. 18. ENOB @ 2.5 MS/s vs. input frequency.



Fig. 19. Microphotograph of the 12-bit 90 nm SAR ADC.

circuits have been included. The A/D converter just occupies an area of about $510 \mu\text{m} \times 600 \mu\text{m}$. The 64 fF metal finger-cap unit capacitors and 100Ω non-silicon P+ doped poly-silicon unit resistors are utilized to form the internal D/A conversion network.

As shown in Fig. 20, the DNL and INL are measured to be less than 0.8 and 2.7 LSB, respectively. In this design, since the MSB is generated based on the two-segment resistor string, the INL of the worst case should be determined by the deviation of the common-mode voltage from its ideal value. The INL of this converter is not well. This is mainly caused by the mismatch of the resistors in the two-segment resistor string, and to overcome this shortcoming is one of our future works. Shown in Fig. 21 is a 4096 points FFT spectrum for a 10 kHz single-ended sine input signal sampled at 1 MS/s. The ENOB and SFDR are measured to be 10.3 and 70.3 dB, respectively. Shown in Fig. 22 is the variation of the ENOB with the input frequency. Including the output drivers, the

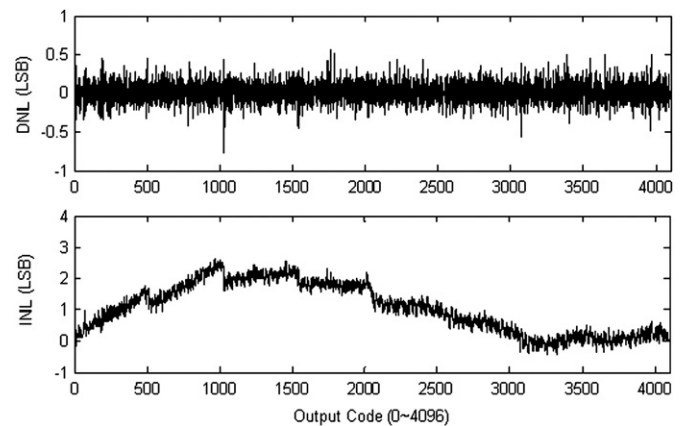


Fig. 20. Nonlinearity of the 12-bit 90 nm SAR ADC.

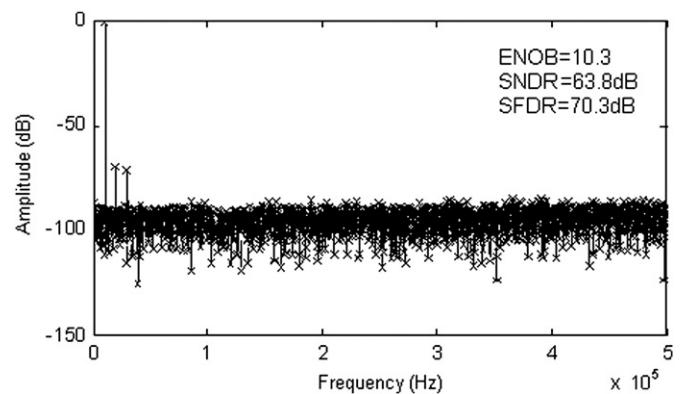


Fig. 21. 4096 points FFT @ 1 MS/s with 10 kHz input.

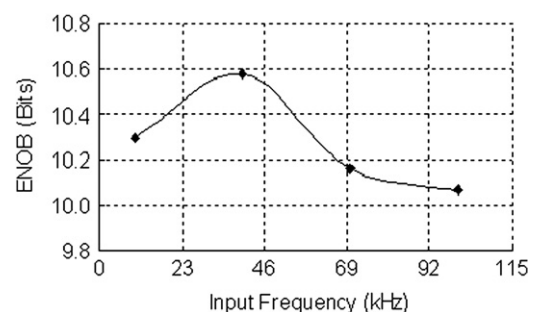


Fig. 22. ENOB @ 1 MS/s vs. input frequency.

Table 1

Comparison with some previous works.

References	Process	Resolution (bit)	Sampling rate (MS/s)	DNL/INL (LSB/LSB)	Power (mW)	Area (mm ²)	ENOB (bit)	FoM (pj/conv. step)
[7]	0.35 μ m	10	2	0.5/0.5	3	0.4	8	5.86
[8]	0.6 μ m	12	1	0.5/0.5	15	1.5	11.6	4.83
[9]	—	12	0.125	< 1.0/2.0	0.65	—	< 12	> 4.6
[14]	90 nm	10	1	—	4.29	0.09	< 10	> 4.2
Proposed	0.18 μ m	10	0.2	0.32/0.8	0.136	0.08	9.15	< 1.2
Proposed	90 nm	10	2.5	0.36/0.69	6.62	0.05	9.41	3.84
Proposed	90 nm	12	1	0.8/2.7	10	0.31	10.3	7.9

power dissipation of this 12-bit 90 nm CMOS SAR ADC is less than 10 mW.

In all the three SAR A/D converters, the comparators are all composed of two preamplifiers with cross-coupled positive feedback [18] and a low power regenerative dynamic latch [4]. Attention must be paid that the tail current of the preamplifiers is a critical design variable for the comparator. Noise, speed and power consumption of the preamplifier are all related to the current. It is important to find an optimal compromise between a small bias current for low power dissipation and a large value to minimize noise and maximize the comparison rate.

3.4. Comparison with previous works

The design results of this three SAR ADC IP cores are summarized and compared with some previous works in Table 1. During the comparison, the FoM (Figure-of-Merit) = $P_{\text{diss}} / (2^{\text{ENOB}} \times f_{\text{sample}})$ is used, where P_{diss} is the power dissipation of the converter, and ENOB is measured at the sampling rate of f_{sample} .

The design and measurement results have proven that these three novel high-resolution D/A conversion networks proposed are applicable to SAR ADCs. Additionally, the comparison with some previous works in Table 1 shows that the three SAR ADCs realized in this paper is characterized by their small area, low power and relatively high SNDR. Especially, profiting from the intermittent-sleeping operation of the resistor string, the 10-bit 0.18 μ m CMOS SAR ADC has the lowest power and the best FoM of all these ADCs in Table 1. The 10-bit 90 nm CMOS SAR ADC features small area, high SNDR and relatively good FoM. Ref. [8] shows a 12-bit SAR ADC with traditional C–R hybrid D/A structure, where high SNDR is achieved due to the fully-differential architecture. But it is not very suitable for the embedded SoC due to its large area (more than four times that of the 12-bit 90 nm SAR ADC in this paper). The three SAR ADC IP cores realized in this paper can satisfy the requirement of the embedded SoC and can be used in touch screen systems, industry control applications and data acquisitions.

4. Conclusion

Three novel hybrid D/A conversion structures are discussed with theoretical analysis, Matlab modeling and circuit realization. The experimental results and the comparison with previous works prove their applicability to high-resolution SAR ADC applications.

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